JINGTAO LEI

Applied Mathematics Undergraduate | AI & Machine Learning Research
Central South University · Dundee International Institute

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2617549@dundee.ac.uk
Website: jingtaolei.github.io
GitHub: jingtaolei

ACADEMIC BACKGROUND

Bachelor of Science (B.Sc.) in Applied Mathematics
Central South University, Dundee International Institute — Changsha, China
Sep 2024 – Jun 2028 (Expected) | Cumulative Average: 87.74/100

ACADEMIC & RESEARCH EXPERIENCE

Position	Organization / Department	Supervisor	Period
SURP Research Student	CUHK, Computer Science & Engineering	Prof. Shengchao Liu	Jun 2026 – Present
Undergraduate Researcher	Multimodal Learning Research Project	Prof. David P. Woodruff	Jan–Jul 2026

SELECTED RESEARCH ACTIVITIES

Video Diffusion Models toward World-Model Capabilities

The Chinese University of Hong Kong — Department of Computer Science and Engineering
Supervisor: Prof. Shengchao Liu | Jun 2026 – Present

- Investigated whether pretrained **video diffusion models encode usable physical dynamics**, with the broader goal of extending visual generative models toward **world-model capabilities** for prediction and physical reasoning.
- Built controlled motion and two-object interaction benchmarks with known physical ground truth and a reproducible **CogVideoX-2B** analysis pipeline; mapped direction, speed, optical flow, and contact information across Transformer layers, diffusion noise levels, and readouts using linear probes, ID/OOD evaluation, and negative controls.
- Found strong linear decodability of motion factors: best controlled ID results reached approximately **1.09° direction error** and **R^2 = 0.988** for speed. The factor map varied across layers, noise levels, and readouts, motivating stronger tests of functional use.
- Designed and initiated **representation-level interventions** using factor-subspace removal, matched random orthogonal controls, and temporal perturbations to distinguish information that is merely decodable from information that is functionally used by the model.

Layer-Wise Gate-Controlled Prompt Truncation in a Multimodal Chest X-Ray Classifier

Undergraduate Research Project
Supervisor: Prof. David P. Woodruff | Jan 2026 – Jul 2026 | arXiv:2609.06590 | Code

- Investigated **gate-controlled prompt shortening** in parameter-efficient multimodal learning, extending a fixed-length prompt framework with a lightweight layer-wise controller.
- Extended a ViT-based **Mixture of Prompt Experts (MoPE)** classifier with retention-ratio predictions; mini-batch-averaged ratios determine the integer retained visual prompt length, while retained mixed prompts are scaled by individual ratios.
- Adapted and evaluated the framework on a derived **IU X-Ray chest X-ray classification task** with controlled gated-versus-fixed prompting experiments.
- In the recorded run, best validation accuracy was **0.8996 vs 0.8969** for the fixed-length baseline and final validation accuracy was **0.8963 vs 0.8802**. Exported gate statistics retained **one of six** visual prompt tokens at all recorded training points; runtime acceleration was not directly measured.

AREAS OF EXPERTISE

Multimodal learning, video diffusion, world models, representation learning, adaptive computation

TECHNICAL SKILLS

Programming & Scientific Computing: Python, Bash, NumPy, pandas, Matplotlib

Machine Learning: PyTorch, Hugging Face Transformers & Diffusers, Vision Transformers, multimodal learning, prompt tuning, diffusion and video diffusion models

Research Methods: controlled dataset construction, representation probing, intermediate-feature analysis, ID/OOD evaluation, negative controls, ablation studies, representation-level interventions

Systems & Research Computing: Linux, SSH-based remote development, Git, Slurm, Singularity, GPU/HPC workflows; hands-on experience running experiments on NVIDIA H200 clusters

PUBLICATIONS

Jingtao Lei, Hongji Li, Dexiang Shu.
Layer-Wise Gate-Controlled Prompt Truncation in a Multimodal Chest X-Ray Classifier.
arXiv preprint arXiv:2609.06590, 2026. | arXiv | Code

AWARDS & ACADEMIC ACHIEVEMENTS

- **Second Prize**, 6th "Celian Cup" Mathematical Modeling Elite League (Provincial-level Competition), 2026
- **First Prize**, 21st Central South University Mathematical Modeling Competition and Internal Selection for the National Competition
- **Third Prize**, Annual Comprehensive Student Evaluation Award, Central South University
- **97.80/100 (A+)**, **From Combinatorics to Calculus**, taught by Prof. Dan Ciubotaru, University of Oxford

ADDITIONAL TRAINING

AWS Academy Graduate — Cloud Foundations